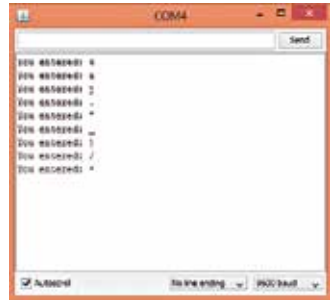


Connect the Arduino™ via USB cable and enter some data in the serial monitor.

For those who need the data in Binary, Hex, Decimal or Octal format, add an extra parameter that specifies the requirement.

<code>Serial.print (data,DEC);</code>	Decimal Format
<code>Serial.print (data,HEX);</code>	Hexadecimal Format
<code>Serial.print (data,OCT);</code>	Octal Format
<code>Serial.print (data,BIN);</code>	Binary Format



Screenshot of Serial Monitor showing serial communication

Getting current position from GPS

Reading data from GPS to get the current location is of prime importance for a location-based application. Writing code for the same is very complicated. Fortunately, there's a library called 'Tiny GPS' that simplifies the process and uses the NMEA (National Marine Electronics Association) specification. The second issue is that the GPS devices use the serial port for communication to get data from the GPS device. There's a library called 'SoftwareSerial' that allows the use of any pin to send and receive serial data. Let's see how to achieve both the tasks.

The software serial library is present in default Arduino™ installation. But the Tiny GPS library can be downloaded from <http://dgit.in/tinygpslib> and needs to be added by going to Sketch > Import Library > Add Library from the menu bar.

You can find a copy of the code at <http://dgit.in/ardgpscode> and related explanation at <http://dgit.in/ardgpsexp>.

Note: Though the Software Serial library is of great use for adding extra serial communication ports, it's computationally expensive from the point of view of processing power. If you're developing applications that are time critical or computationally heavy or that don't function as required when using the Software Serial library, it's better to avoid using the library. Better to use an Arduino™ board with multiple serial communication ports such as the Arduino™ Mega 2560 or the Arduino™ Mega ADK.

Connecting an LCD display

An LCD display is great for getting instant feedback on what the Arduino™